

Detecting Structural Breaks in Crypto Markets

A beginner-friendly empirical report inspired by structural-break methods in
Advances in Financial Machine Learning

Prepared from Binance daily OHLC market data

Generated on 2026-05-19

What this report does. We take historical crypto price data for BTC, ETH, ETC, SOL, and HYPE and apply structural-break diagnostics: a BDE-inspired recursive forecast-error CUSUM, Chu-Stinchcombe-White CUSUM, Chow-type Dickey-Fuller, SADP, QADF, a conditional-tail ADF summary, and sub/super-martingale trend tests. The goal is educational: show how the tests behave, how to interpret the plots, and why one should treat flagged regimes as research features rather than automatic trading signals.

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1 Intuition

A **structural break** is a change in the data-generating mechanism. In markets this often means that a process that looked stable, noisy, or mean-reverting starts behaving like a momentum process, an explosive bubble, or a fast collapse. These breaks can be useful because many market participants adapt slowly, but the diagnostics are not crash predictors. A strong reading says: *something about the recent price path is unusual under the reference model*. That unusual behavior may be a bubble, a burst, a volatility shock, a liquidity episode, a listing effect, a leverage unwind, or a false alarm.

2 Data and Parameters

The raw source data are Binance daily OHLCV klines. The report uses the largest available Binance spot USDT history where spot is available. HYPEUSDT is not available on Binance spot in the API check used here, so HYPE falls back to Binance USD-M futures daily klines. All diagnostics are computed on daily closes through 2026-05-18.

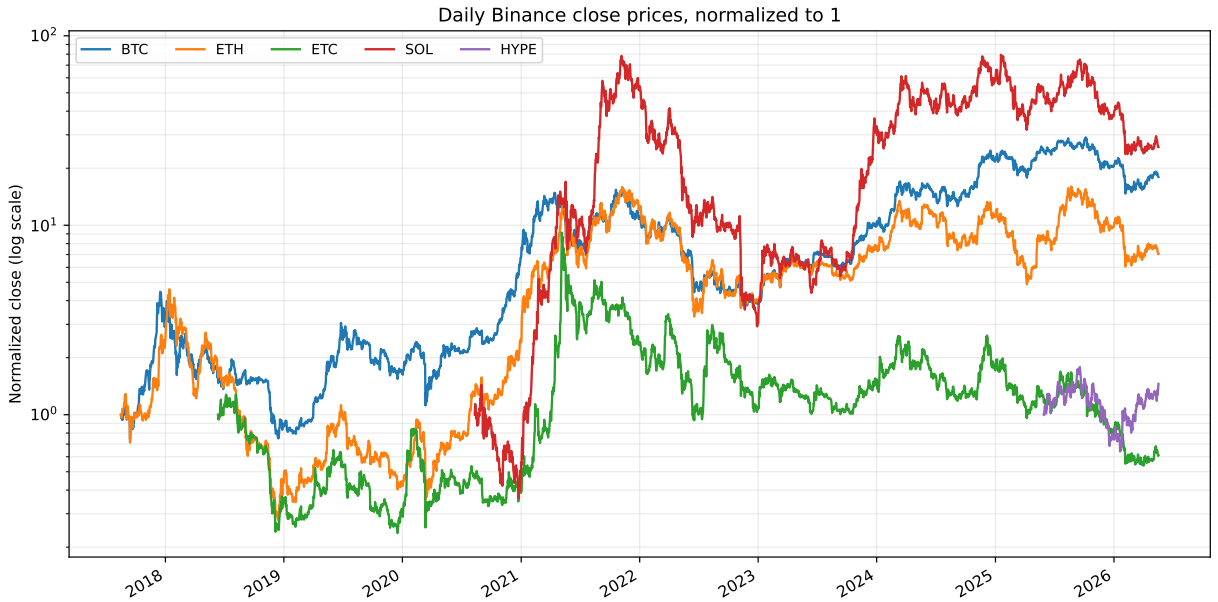


Figure 1: Downloaded daily close prices, normalized to 1 at each asset’s first available row. A log scale is used because crypto prices can move by orders of magnitude.

Asset	Source	Raw start	Raw end	Raw rows	τ	SDFC candidate	SDFC t	Max SADF	Max CSW	Max SMT	Runs
BTC	Binance spot	2017-08-17	2026-05-18	3197	480	2018-12-16	1.51	2021-01-08	2020-03-12	2025-10-09	16
ETH	Binance spot	2017-08-17	2026-05-18	3197	480	2020-03-13	0.99	2021-05-11	2020-03-12	2021-12-02	20
ETC	Binance spot	2018-06-12	2026-05-18	2898	435	none (non-positive t)	-0.19	2021-05-06	2020-03-12	2021-11-03	5
SOL	Binance spot	2020-08-11	2026-05-18	2107	317	2022-12-30	0.62	2023-12-25	2022-11-09	2023-01-12	9
HYPE	Binance USD-M futures	2025-05-30	2026-05-18	354	60	2026-01-21	1.44	2025-12-18	2026-01-28	2025-12-28	5

Table 1: Daily Binance data and headline diagnostic results. τ is the minimum daily window length used in recursive tests. Runs count retrospective consensus alert runs, where at least two diagnostic families were simultaneously in their research-alert region.

Parameter	Value
Transaction cost	0.10% per exposure change
Online signal quantile	0.95 prior expanding quantile
Minimum online history	60 finite observations
Momentum gate	trailing 20-day log return must be positive
Holding window	20 calendar daily observations after an alert
Minimum diagnostic window	$\max(60, \lceil 0.15n \rceil)$
ADF Monte Carlo calibration	2000 random-walk simulations, seed 123
Walk-forward validation	expanding 730-day train, 180-day test blocks

Table 2: Backtest and diagnostic parameters used by the generated report.

3 Diagnostic Families

3.1 CUSUM Tests

CUSUM means cumulative sum. The BDE-inspired series in this report is a recursive one-step forecast-error CUSUM from a simple log-price trend model. It is **not** a formal Brown-Durbin-Evans boundary test: residuals are normalized online with shifted expanding statistics so a value at close t uses only information available through t .

The Chu-Stinchcombe-White statistic asks whether log price has moved too far from an earlier reference level relative to realized volatility:

$$S_{n,t} = \frac{y_t - y_n}{\hat{\sigma}_t \sqrt{t - n}}, \quad \hat{\sigma}_t^2 = \frac{1}{t - 1} \sum_{i=2}^t (\Delta y_i)^2.$$

The report applies the level-departure statistic symmetrically to bubbles and crashes. Because of that absolute-value transformation, the displayed CSW excess should be read as a two-sided heuristic rather than the original one-sided 5% test.

3.2 ADF-Style Tests

SDFC scans for a single full-sample unknown break into an explosive alternative. It is not traded here because choosing the break date from the full sample would use future data. SADF, QADF, and CADF scan repeated backward windows ending at each date. The implementation uses the zero-lag ADF form for readability and speed:

$$\Delta y_t = \alpha + \beta y_{t-1} + \varepsilon_t.$$

For formal interpretation, the report now adds cached finite-sample random-walk simulations for SADF/QADF/CADF critical values. The full-sample top 5% plot flags remain research visualizations, not p-values.

3.3 SMT-Style Trend Tests

The sub/super-martingale score evaluates exponential and quadratic trend forms on normalized price and log price. High values flag unusually strong trend-like growth or collapse. They are useful features, but the reported score is a compact heuristic rather than a complete production SMT battery.

4 Formal Inference Versus Research Features

Statistic	Threshold type	Formal p-value?	Online?
BDE-inspired CUSUM	shifted expanding residual z-score	no	yes for strategy, no for retrospective plot flags
Two-sided CSW excess	book-inspired curve applied symmetrically	heuristic	yes
SADF/QADF/CADF visual flags	full-sample top 5%	no	no
SADF/QADF/CADF critical exceedances	simulated random-walk finite-sample critical values	approximate	no, retrospective inference only
SDFC	full-sample unknown-break scan	no in this report	no
Long/flat strategies	prior expanding thresholds plus one-day execution shift	no	yes

Table 3: Interpretation labels for the report’s diagnostics and strategy conversions.

5 How to Read the Plots

Each asset figure has five panels: price, BDE-inspired recursive CUSUM, two-sided CSW excess, ADF-family statistics, and SMT-style trend score. Orange shading marks **retrospective consensus alert periods**: at least two diagnostic families were in their full-sample top-alert region or exceeded the fixed CSW boundary. This shading is useful for visual structure clustering, but it is not an online trading signal.

The online strategy signals are separate. BDE, SADF, QADF, CADF, and SMT strategy events use prior expanding thresholds only. CSW strategy events use the current positive excess because the statistic already embeds its reference boundary.

6 Results by Asset

6.1 BTC

16 retrospective consensus alert run(s) were flagged on the daily clock. The first listed run begins on 2021-01-02. The SDFC candidate is 2018-12-16 with t-statistic 1.51. SADF peaks on 2021-01-08, the two-sided CSW heuristic peaks on 2020-03-12, and the SMT score peaks on 2025-10-09.

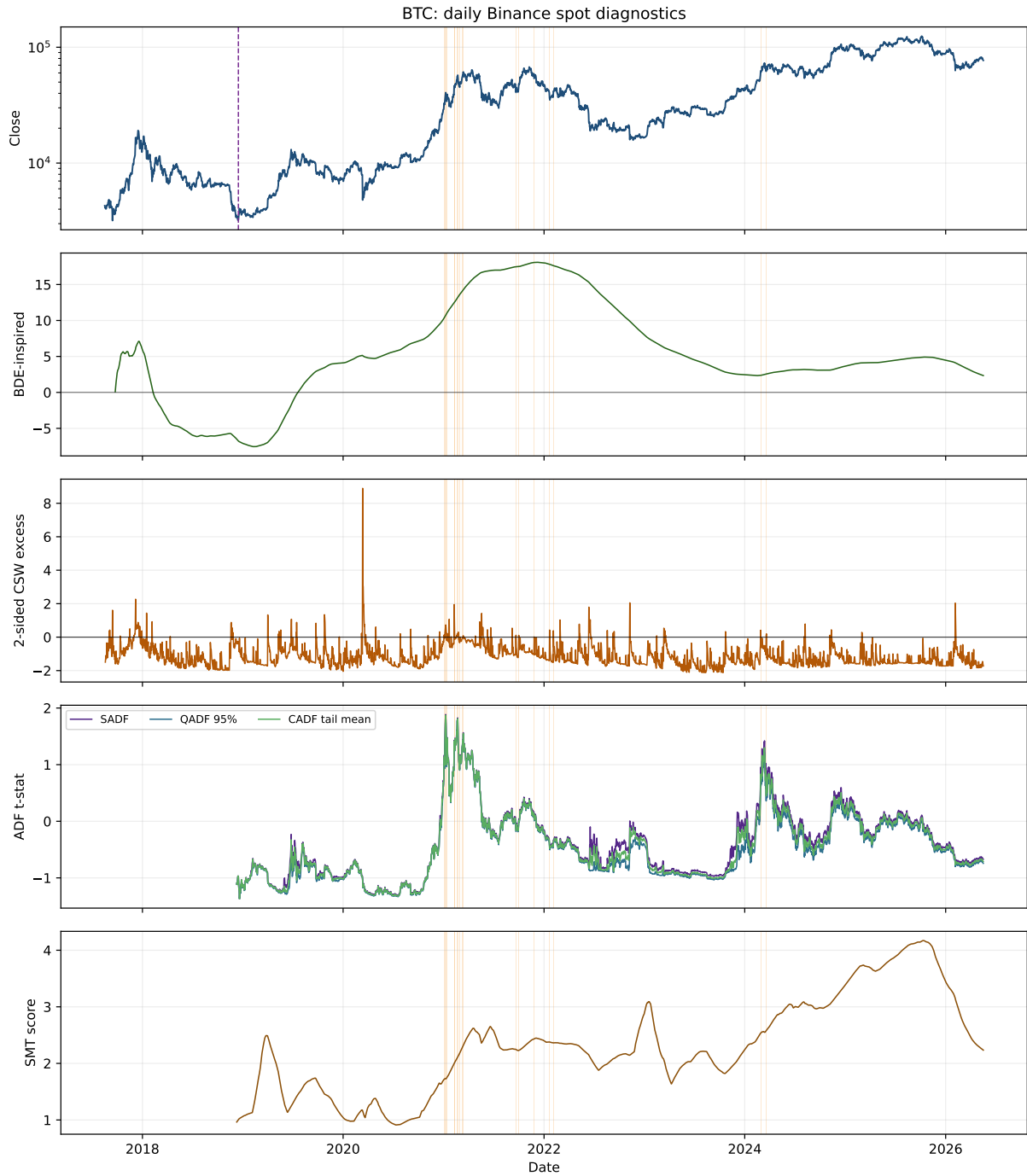


Figure 2: BTC structural-break diagnostics. Orange shading is retrospective consensus, not an online trading signal.

6.2 ETH

20 retrospective consensus alert run(s) were flagged on the daily clock. The first listed run begins on 2021-01-04. The SDFC candidate is 2020-03-13 with t-statistic 0.99. SADF peaks on 2021-05-11, the two-sided CSW heuristic peaks on 2020-03-12, and the SMT score peaks on 2021-12-02.

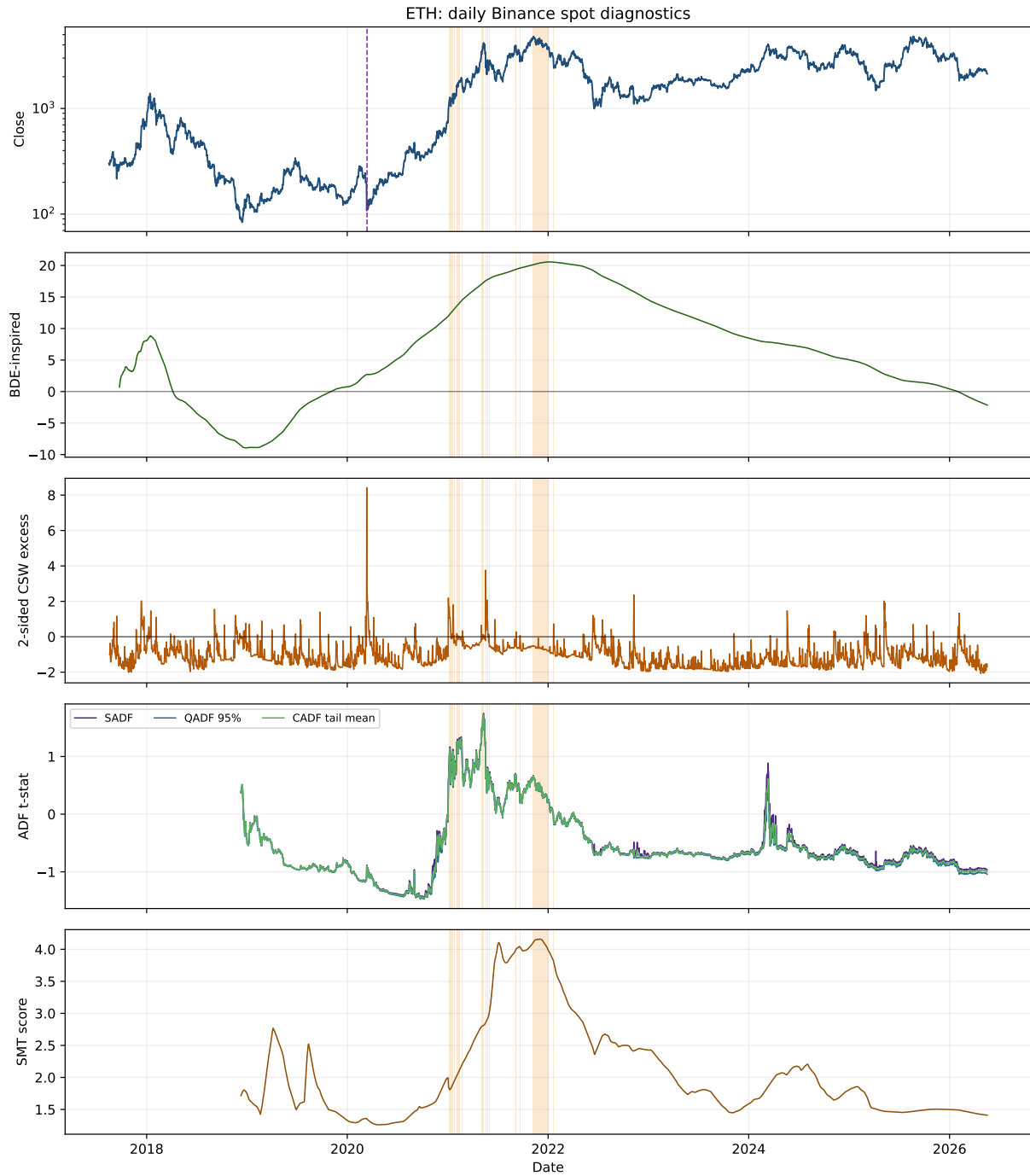


Figure 3: ETH structural-break diagnostics. Orange shading is retrospective consensus, not an online trading signal.

6.3 ETC

5 retrospective consensus alert run(s) were flagged on the daily clock. The first listed run begins on 2021-04-15. The SDFC candidate is none (non-positive t) with t -statistic -0.19. SADF peaks on 2021-05-06, the two-sided CSW heuristic peaks on 2020-03-12, and the SMT score peaks on 2021-11-03.

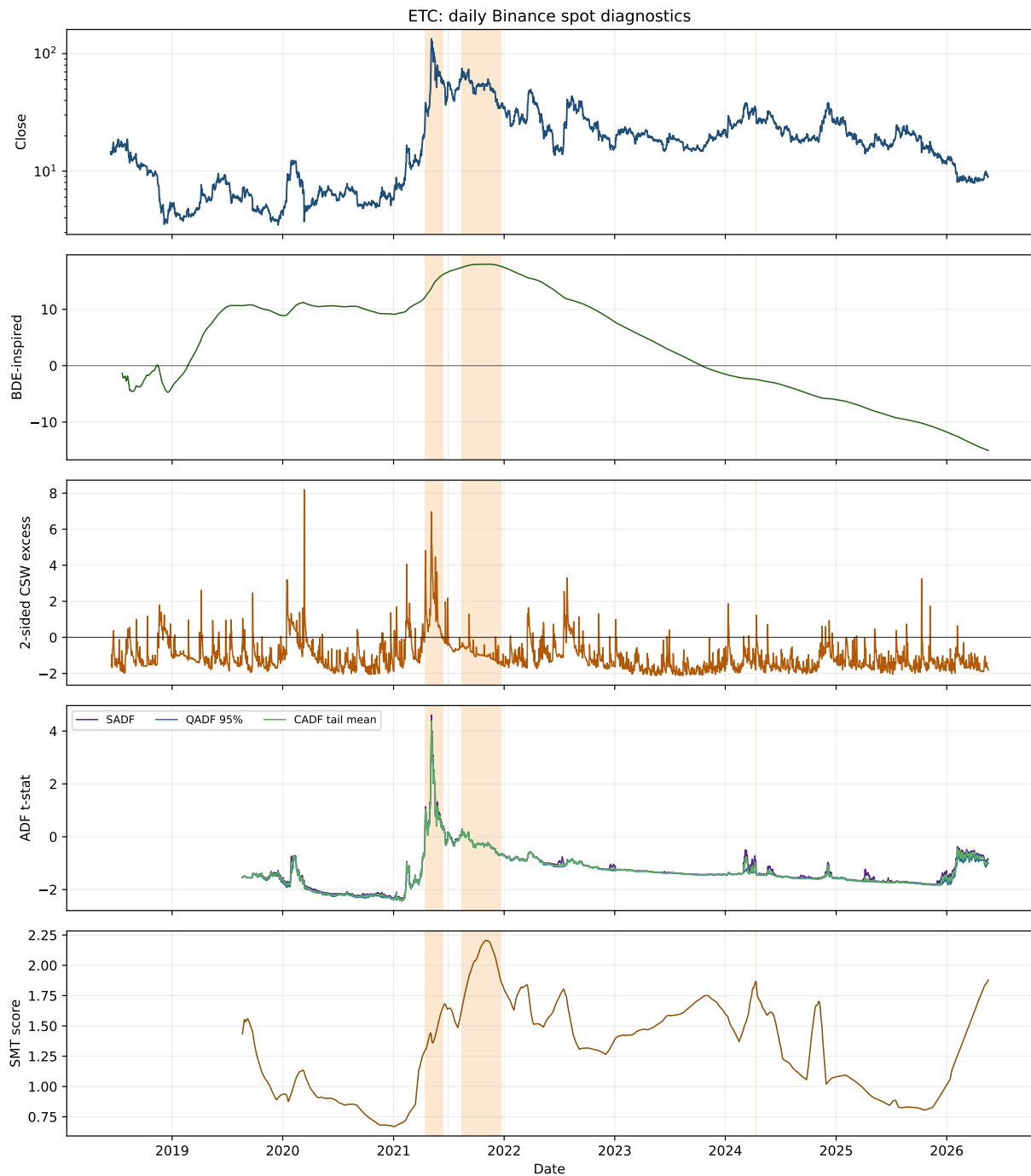


Figure 4: ETC structural-break diagnostics. Orange shading is retrospective consensus, not an online trading signal.

6.4 SOL

9 retrospective consensus alert run(s) were flagged on the daily clock. The first listed run begins on 2021-09-07. The SDFC candidate is 2022-12-30 with t-statistic 0.62. SADF peaks on 2023-12-25, the two-sided CSW heuristic peaks on 2022-11-09, and the SMT score peaks on 2023-01-12.

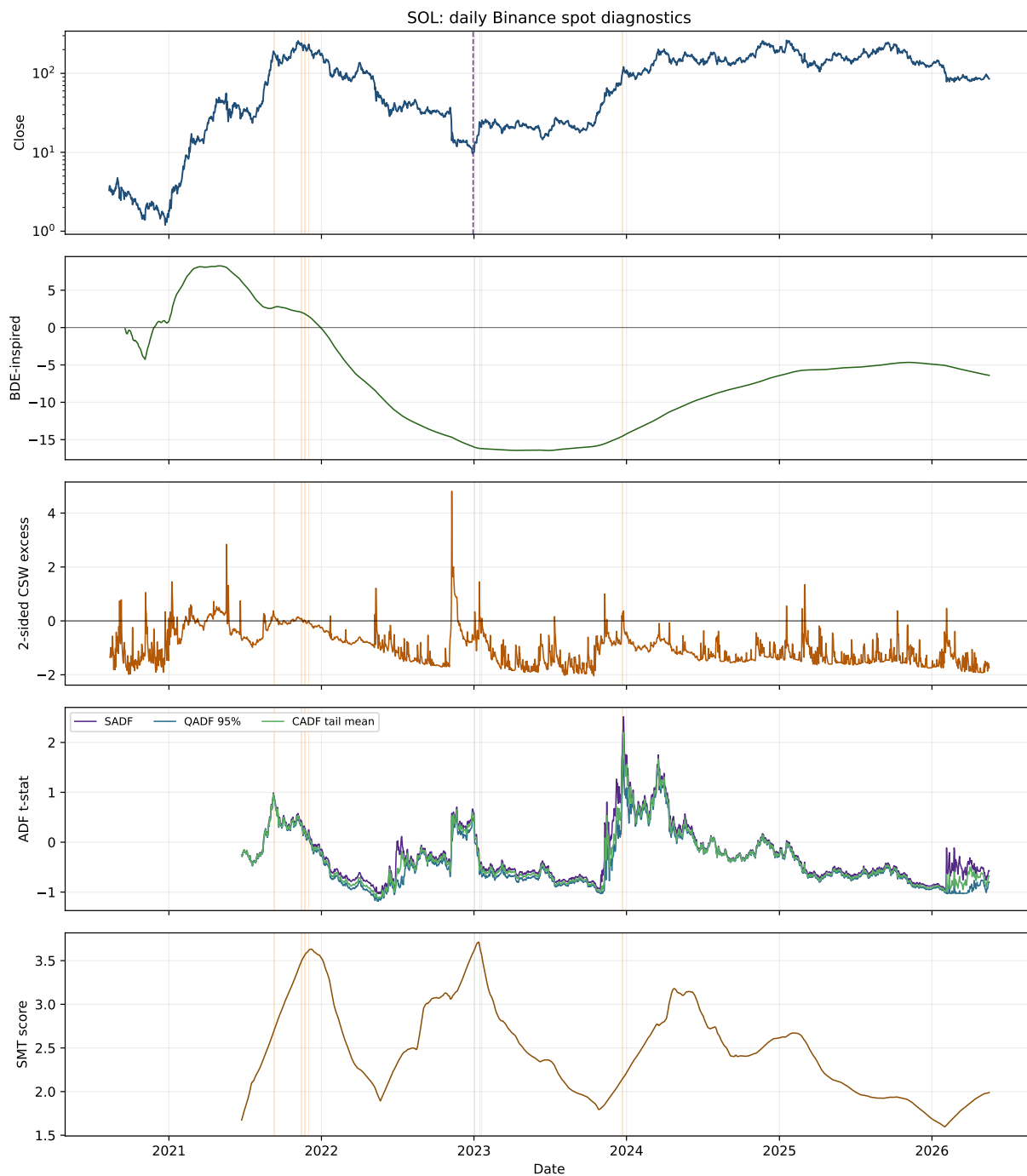


Figure 5: SOL structural-break diagnostics. Orange shading is retrospective consensus, not an online trading signal.

6.5 HYPE

5 retrospective consensus alert run(s) were flagged on the daily clock. The first listed run begins on 2025-12-18. The SDFC candidate is 2026-01-21 with t-statistic 1.44. SADF peaks on 2025-12-18, the two-sided CSW heuristic peaks on 2026-01-28, and the SMT score peaks on 2025-12-28. HYPE uses Binance USD-M futures price klines; strategy returns exclude funding, carry, basis, and futures-specific execution costs.

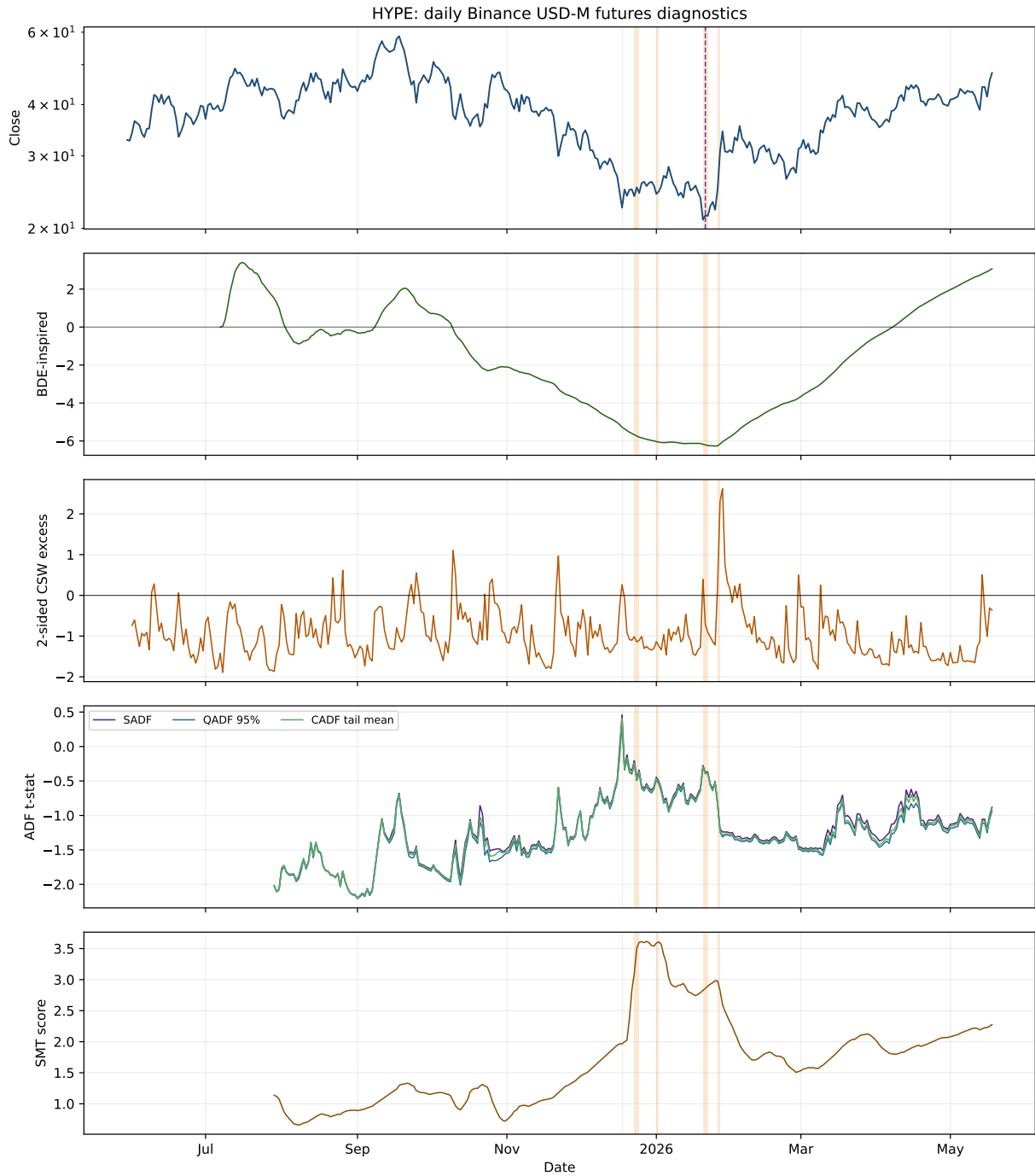


Figure 6: HYPE structural-break diagnostics. Orange shading is retrospective consensus, not an online trading signal.

7 Consensus Alert Periods

The table below lists the first few retrospective periods in which at least two diagnostic families were active. Short alert runs are common because the tests are sensitive to local trend acceleration and volatility normalization.

Table 4: First listed retrospective consensus alert periods. Period length is counted in calendar days between the first and last daily alert observation. For assets with many daily runs, the table shows the first eight runs.

Asset	Start	End	Days/period length
BTC	2021-01-02	2021-01-03	2
	2021-01-05	2021-01-10	6
	2021-01-13	2021-01-14	2
	2021-02-08	2021-02-09	2
	2021-02-11	2021-02-11	1
	2021-02-17	2021-02-23	7
	2021-03-01	2021-03-01	1
	2021-03-11	2021-03-11	1
ETH	2021-01-04	2021-01-10	7
	2021-01-14	2021-01-14	1
	2021-01-16	2021-01-16	1
	2021-01-19	2021-01-20	2
	2021-01-24	2021-01-24	1
	2021-02-03	2021-02-03	1
	2021-02-05	2021-02-05	1
	2021-02-08	2021-02-09	2
ETC	2021-04-15	2021-06-11	58
	2021-06-13	2021-06-14	2
	2021-06-29	2021-06-30	2
	2021-08-14	2021-12-23	132
	2024-04-12	2024-04-12	1
SOL	2021-09-07	2021-09-11	5
	2021-11-13	2021-11-17	5
	2021-11-20	2021-11-21	2
	2021-11-23	2021-11-23	1
	2021-12-01	2021-12-02	2
	2023-01-03	2023-01-03	1
	2023-01-14	2023-01-16	3
	2023-01-20	2023-01-20	1
HYPE	2025-12-18	2025-12-18	1
	2025-12-23	2025-12-25	3
	2026-01-01	2026-01-02	2
	2026-01-20	2026-01-22	3
	2026-01-26	2026-01-27	2

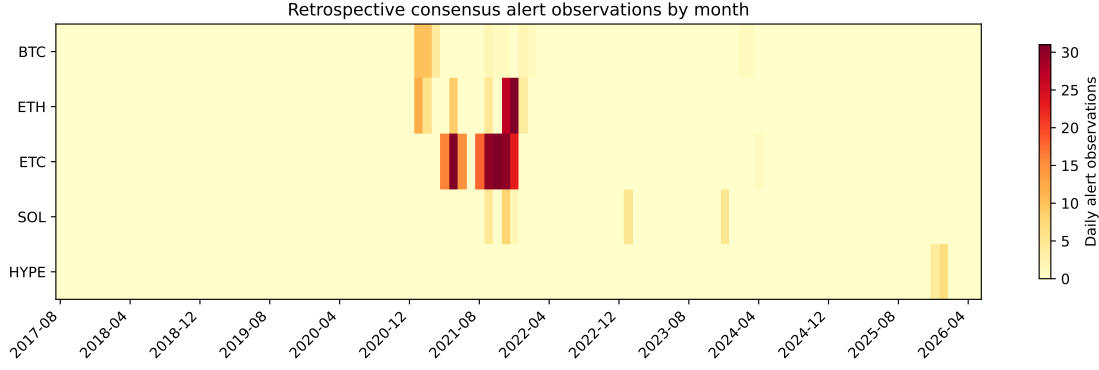


Figure 7: Retrospective consensus alert observations by month.

8 Finite-Sample Calibration and Sensitivity

The table below compares each asset’s observed ADF-family maxima with cached random-walk finite-sample critical values. These simulations give a more defensible reference than the internal top 5% visual flags, although they still inherit the simplified zero-lag ADF specification.

Asset	Sims	τ	SADF 95%	Max SADF	SADF hits	QADF 95%	Max QADF	QADF hits	CADF 95%	Max CADF	CADF hits
BTC	2000	480	2.01	1.89	0	1.93	1.82	0	1.96	1.87	0
ETH	2000	480	2.01	1.75	0	1.93	1.72	0	1.96	1.73	0
ETC	2000	435	2.02	4.61	13	1.91	4.29	14	1.96	4.38	13
SOL	2000	317	2.02	2.52	4	1.91	1.67	0	1.96	2.20	1
HYPE	2000	60	1.98	0.46	0	1.89	0.32	0	1.94	0.41	0

Table 5: Simulated finite-sample critical values under random-walk null paths with matching sample length and τ . Hits count daily observations above the 95% simulated max-statistic threshold.

Table 6: Sensitivity of selected results to the minimum-window fraction. ADF-family windows vary; BDE, CSW, and SMT inputs are held at their main-run definitions.

Asset	τ rule	τ	Max SADF date	Runs	Best structure signal	Final
BTC	10%	320	2024-03-13	18	CSW	11.68x
BTC	15%	480	2021-01-08	16	CSW	11.68x
BTC	20%	640	2021-02-21	15	CSW	11.68x
ETH	10%	320	2024-03-11	23	BDE	9.35x
ETH	15%	480	2021-05-11	20	BDE	9.35x
ETH	20%	640	2021-05-11	20	Consensus	11.54x
ETC	10%	290	2021-05-06	13	CSW	11.44x
ETC	15%	435	2021-05-06	5	CSW	11.44x
ETC	20%	580	2021-05-06	5	CSW	11.44x
SOL	10%	211	2023-12-25	17	CSW	27.43x
SOL	15%	317	2023-12-25	9	CSW	27.43x
SOL	20%	422	2024-03-17	14	CSW	27.43x
HYPE	10%	60	2025-12-18	5	BDE	1.10x
HYPE	15%	60	2025-12-18	5	BDE	1.10x
HYPE	20%	71	2025-12-18	5	BDE	1.10x

9 Long/Flat Backtests

The diagnostics above are research features, not trading rules. Still, it is useful to ask what a simple long-or-cash conversion would have done.

9.1 Backtest Rule

1. At each daily close t , compute diagnostics using data through t only.
2. Convert each eligible indicator into an online alert using prior expanding thresholds, except CSW, whose positive excess is already boundary-based.
3. Require the trailing momentum gate to be positive.
4. An alert computed at close t sets exposure for the close t to close $t + 1$ return, assuming execution at or immediately after close t .
5. Keep the position long for the holding window after a valid alert, then move back out unless a new alert refreshes the holding window.
6. Charge transaction costs whenever exposure changes. Buy-and-hold is charged one entry cost for consistency.

This is still an in-sample comparison. The best final PnL is an indicative ranking of which simple signal conversion looked promising on this sample, not evidence of a deployable strategy.

Asset	Best structure signal	Best signal final	Buy-and-hold final	Best including buy-and-hold	Max DD	Exposure	Trades
BTC	CSW	11.68x	17.95x	Buy-and-hold	-37%	17%	20
ETH	BDE	9.35x	7.05x	BDE	-61%	19%	7
ETC	CSW	11.44x	0.61x	CSW	-67%	26%	24
SOL	CSW	27.43x	25.86x	CSW	-60%	21%	9
HYPE	BDE	1.10x	1.45x	Buy-and-hold	-26%	12%	1

Table 7: Best long/flat structure-break strategy by final equity multiple. Best structure signal excludes buy-and-hold; best including buy-and-hold includes it. Max DD, exposure, and trades refer to the best structure signal.

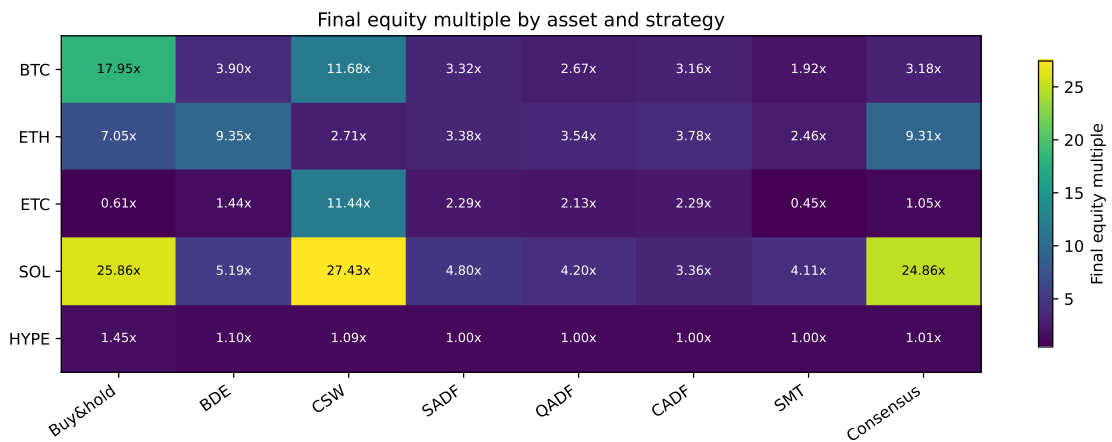


Figure 8: Final equity multiple by asset and strategy. Values are in-sample and normalized to 1 at each asset's first available observation.

9.2 PnL Curves

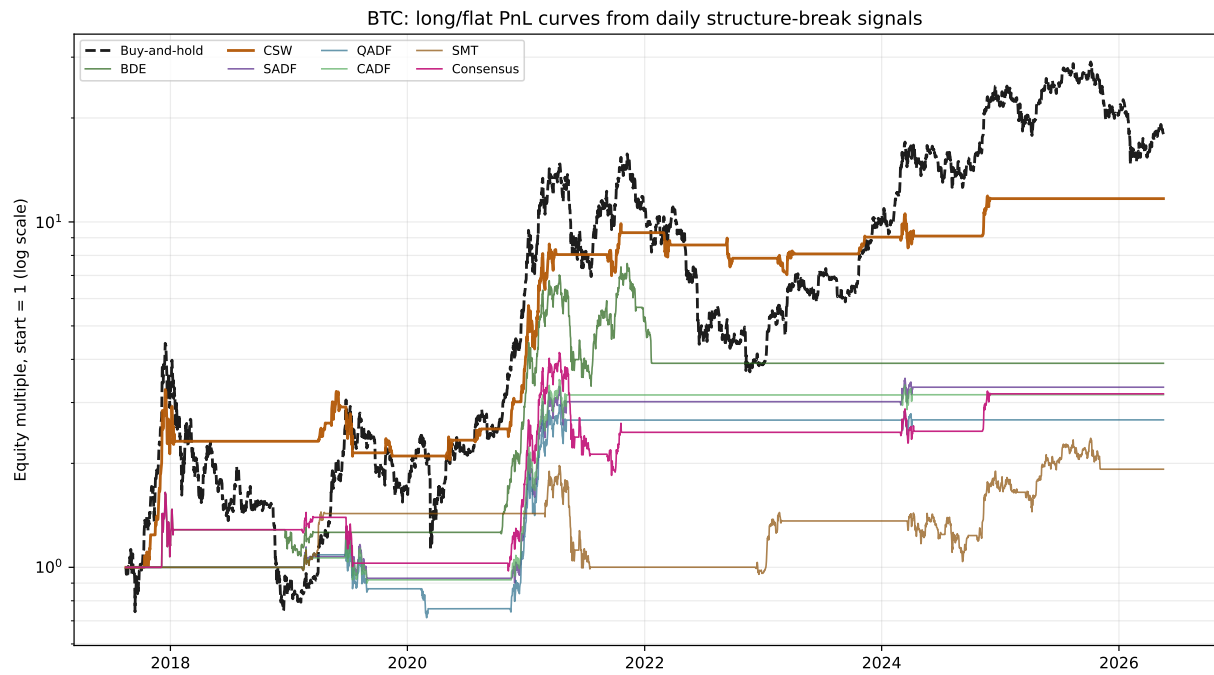


Figure 9: BTC long/flat strategy PnL curves.

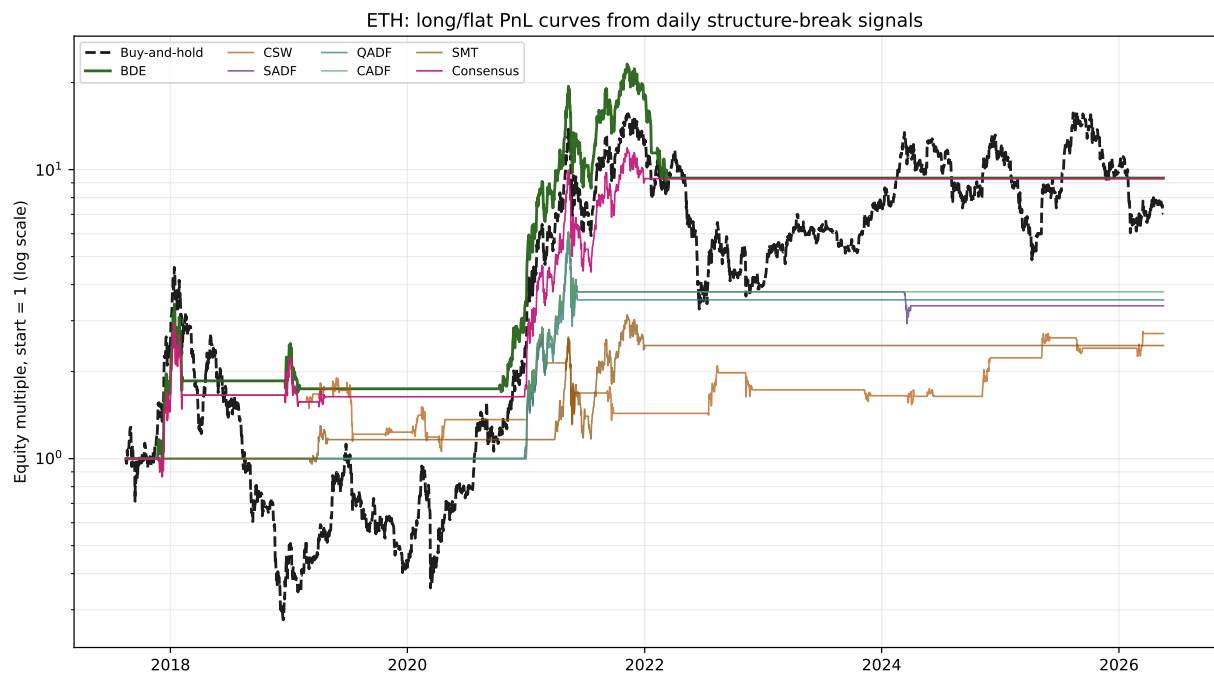


Figure 10: ETH long/flat strategy PnL curves.



Figure 11: ETC long/flat strategy PnL curves.

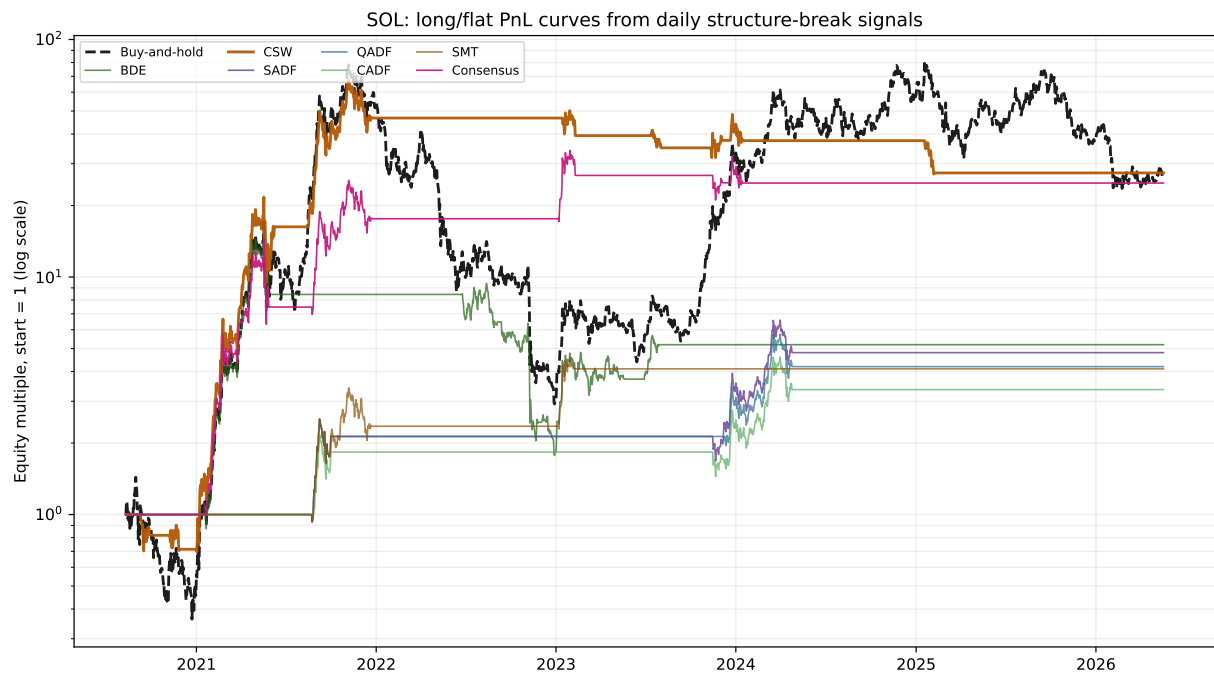


Figure 12: SOL long/flat strategy PnL curves.

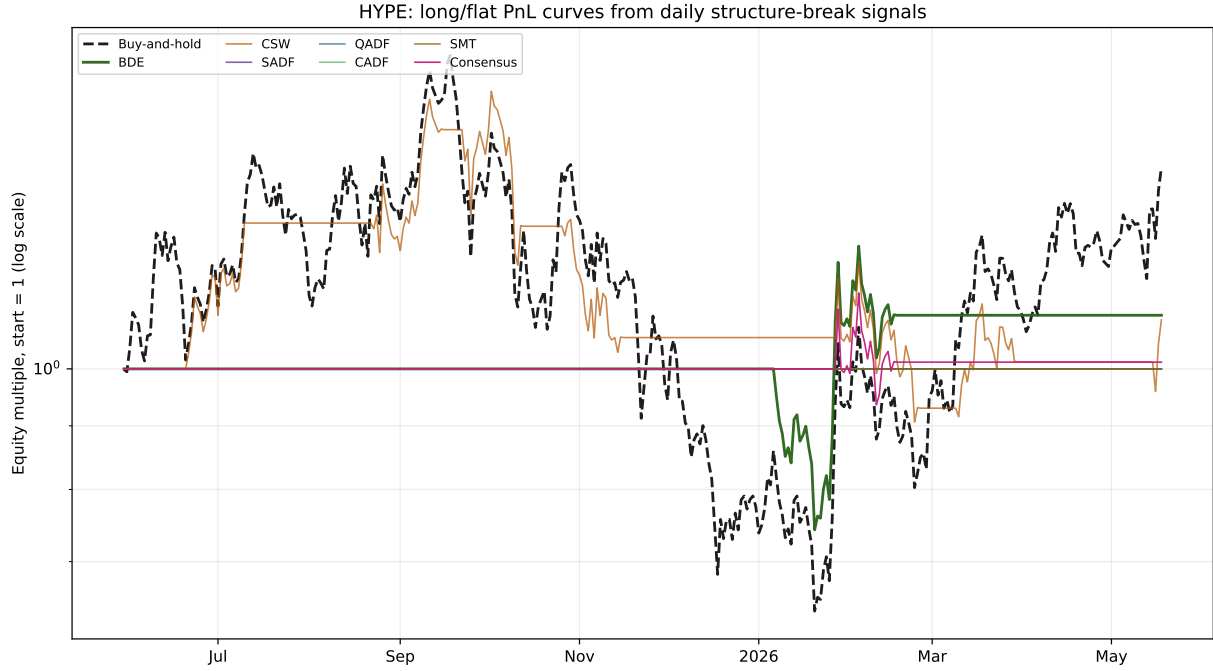


Figure 13: HYPE long/flat strategy PnL curves. HYPE uses Binance USD-M futures price data; returns exclude funding and futures-specific carry.

9.3 Full Strategy Table

Table 8: Simple long/flat strategy performance. Final multiple is the ending equity multiple from 1. CAGR is annualized over each asset’s available sample. Sharpe uses daily returns and 365-day annualization.

Asset	Strategy	Final	Total return	CAGR	Max DD	Sharpe	Exposure	Trades
BTC	Buy-and-hold	17.95x	1695%	39%	-83%	0.83	100%	1
BTC	BDE	3.90x	290%	17%	-52%	0.62	17%	6
BTC	CSW	11.68x	1068%	32%	-37%	1.00	17%	20
BTC	SADF	3.32x	232%	15%	-31%	0.66	9%	5
BTC	QADF	2.67x	167%	12%	-41%	0.56	11%	6
BTC	CADF	3.16x	216%	14%	-30%	0.64	9%	5
BTC	SMT	1.92x	92%	8%	-52%	0.41	25%	11
BTC	Consensus	3.18x	218%	14%	-56%	0.56	14%	11
ETH	Buy-and-hold	7.05x	605%	25%	-94%	0.70	100%	1
ETH	BDE	9.35x	835%	29%	-61%	0.77	19%	7
ETH	CSW	2.71x	171%	12%	-64%	0.48	18%	22
ETH	SADF	3.38x	238%	15%	-52%	0.61	6%	2
ETH	QADF	3.54x	254%	16%	-50%	0.63	5%	1
ETH	CADF	3.78x	278%	16%	-50%	0.66	5%	1
ETH	SMT	2.46x	146%	11%	-56%	0.49	10%	4
ETH	Consensus	9.31x	831%	29%	-56%	0.78	15%	6
ETC	Buy-and-hold	0.61x	-39%	-6%	-94%	0.43	100%	1
ETC	BDE	1.44x	44%	5%	-74%	0.35	16%	7
ETC	CSW	11.44x	1044%	36%	-67%	0.77	26%	24
ETC	SADF	2.29x	129%	11%	-61%	0.44	5%	3
ETC	QADF	2.13x	113%	10%	-61%	0.42	5%	3
ETC	CADF	2.29x	129%	11%	-61%	0.44	5%	3

Asset	Strategy	Final	Total return	CAGR	Max DD	Sharpe	Exposure	Trades
ETC	SMT	0.45x	-55%	-9%	-67%	-0.19	6%	5
ETC	Consensus	1.05x	5%	1%	-79%	0.27	13%	10
SOL	Buy-and-hold	25.86x	2486%	76%	-96%	1.06	100%	1
SOL	BDE	5.19x	419%	33%	-89%	0.76	23%	6
SOL	CSW	27.43x	2643%	78%	-60%	1.16	21%	9
SOL	SADF	4.80x	380%	31%	-35%	0.94	9%	2
SOL	QADF	4.20x	320%	28%	-35%	0.91	8%	2
SOL	CADF	3.36x	236%	23%	-35%	0.78	9%	3
SOL	SMT	4.11x	311%	28%	-40%	0.85	7%	2
SOL	Consensus	24.86x	2386%	75%	-56%	1.20	15%	6
HYPE	Buy-and-hold	1.45x	45%	47%	-64%	0.87	100%	1
HYPE	BDE	1.10x	10%	11%	-26%	0.44	12%	1
HYPE	CSW	1.09x	9%	10%	-46%	0.45	38%	7
HYPE	SADF	1.00x	0%	0%	0%	n/a	0%	0
HYPE	QADF	1.00x	0%	0%	0%	n/a	0%	0
HYPE	CADF	1.00x	0%	0%	0%	n/a	0%	0
HYPE	SMT	1.00x	0%	0%	0%	n/a	0%	0
HYPE	Consensus	1.01x	1%	1%	-19%	0.18	6%	1

9.4 Walk-Forward and Multiple-Testing Checks

The walk-forward check selects the best structure signal on an expanding training window and evaluates only the next out-of-sample block. This does not prove profitability, but it reduces the most obvious winner-picking bias in the in-sample table.

Asset	Blocks	OOS days	Selected signal path	OOS final	OOS Sharpe	OOS Max DD
BTC	13	2340	CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW	5.45x	1.08	-29%
ETH	13	2340	BDE, BDE, BDE, BDE, BDE, BDE, BDE, BDE, BDE, BDE, BDE, BDE, BDE, BDE	5.36x	0.79	-61%
ETC	12	2160	CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW	4.58x	0.69	-67%
SOL	7	1260	CSW, CSW, CSW, CSW, CSW, CSW, CSW, CSW	0.59x	-0.35	-46%
HYPE	0	0		n/a	n/a	n/a

Table 9: Expanding-window walk-forward validation of signal selection. HYPE has too little history for the default train/test split and may show no blocks.

Asset	Bootstrap sims	Best signal	Best final	Final 5–95% CI	Sharpe 5–95% CI	Reality-check p
BTC	500	CSW	11.68x	[2.33x, 87.12x]	[0.48, 1.59]	0.956
ETH	500	BDE	9.35x	[1.02x, 138.62x]	[0.24, 1.34]	0.918
ETC	500	CSW	11.44x	[0.55x, 378.68x]	[0.13, 1.35]	0.590
SOL	500	CSW	27.43x	[1.56x, 759.32x]	[0.45, 1.86]	0.958
HYPE	500	BDE	1.10x	[0.67x, 1.97x]	[-1.57, 1.46]	0.898

Table 10: Block-bootstrap uncertainty for the best in-sample structure signal and a simple centered bootstrap reality-check p -value for best excess mean return versus buy-and-hold.

10 Interpretation

Methodologically, SADF is a strong default diagnostic for repeated explosive episodes because it does not require the analyst to know the number or timing of regime switches. Empirically, in this particular in-sample long/flat conversion, the winning signal can differ by asset and must be treated as selected after looking at many alternatives.

The most important traps are multiple testing, listing-history differences, simplified zero-lag ADF regressions, and price-only data. HYPE is especially limited because it uses futures price klines without funding-rate adjustment. Regime-change diagnostics are best read as feature-engineering tools, not as a recipe for direct trading.

11 Minimal Reproducibility Sketch

```
# online execution convention used in the backtest
event_t = signal_at_close_t
position = event_t.rolling(hold_days).max().shift(1)
strategy_return_t = position_t * close.pct_change_t - costs_t
```

12 Reproduce

```
python3 scripts/generate_daily_regime_chapter.py --use-cache
pdflatex -interaction=nonstopmode regime_detection_crypto_report.tex
pdflatex -interaction=nonstopmode regime_detection_crypto_report.tex
```

The generator writes daily diagnostics, strategy PnL CSVs, cached calibration files, strict JSON metadata, all figures, and this TeX report. The rendered PDF is then produced by `pdflatex`.

13 References

- Marcos López de Prado, *Advances in Financial Machine Learning*, Wiley, 2018, structural-break methods.
- Binance API daily kline endpoints for BTCUSDT, ETHUSDT, ETCUSDT, SOLUSDT spot, and HYPEUSDT USD-M futures fallback.
- Local reproduction script: `scripts/generate_daily_regime_chapter.py`.